

The Perils of Bilateral Sovereign Debt*

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Abstract

We study the interaction between private and official sovereign debts. We develop a quantitative sovereign default model featuring a senior lender with whom borrowing terms are negotiated. We use this model to evaluate implications of the emergence of new official creditors not bound by the Paris Club framework. The dynamics of bilateral bargaining lead the government to issue more market debt, raising default risk and creating welfare losses. This relational overborrowing effect arises due to an endogenous cross-elasticity of bilateral terms to market debt, which can be assessed in practice to evaluate new forms of bilateral sovereign debt.

JEL Classification F34, F41, G15

Keywords Sovereign debt, debt dilution, bilateral bargaining, official debt

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INTRODUCTION

A large fraction of sovereign borrowing in emerging-market economies takes the form of official debt, including loans from other governments, regional development banks, or multilateral institutions such as the IMF and the World Bank. The past few decades have witnessed the rise of new sovereign creditors operating outside of the Paris Club institutional framework (Horn et al., 2021b; Gelpern et al., 2021). The claims to de facto senior creditor status (relative to private creditors) by these new lenders have raised concerns about welfare implications for borrowing countries (Horn et al., 2021a, 2023).

This paper evaluates such concerns in the context of a quantitative sovereign debt and default model, augmented by the presence of a large lender who offers an alternative source of funds. Relative to creditors in international capital markets, the large lender has a superior enforcement technology. This assumption allows us to model the potential success of the new sovereign creditors in becoming senior to bondholders in capital markets. In addition, in our baseline setup, we assume that borrowing terms with the large lender are determined through bargaining, which gives a crucial role to the dynamics of the bilateral relationship.

We focus on the interaction between private markets and the large lender as two potential sources of financing.¹ While the availability of bilateral loans from the large lender affects the government's behavior in private debt markets, outcomes in these markets also influence threat points in the bilateral negotiation. The interest rate charged by the large lender is therefore constrained by implicit competition from debt markets. But when default risk pushes up yields on marketable debt, the large lender is also able to charge a premium. Because there is no default risk on the bilateral loan, such a premium only reflects the borrower's outside options.

The baseline model yields two main results: the presence of the senior lender leads to a form of overborrowing in debt markets, and this creates welfare losses for the government. While bilateral loans can and are often used on the equilibrium path to avoid costly defaults, the borrowing country is worse off when the large lender is present. One reason for this is that the possibility of borrowing from the large lender while being excluded from private markets raises the value of default, which, in turn, increases default incentives and lowers (marketable) debt prices. But there is another, more fundamental reason for the welfare losses. Even imposing the constraint that bilateral loans are unavailable during default, welfare for the government is still reduced by the presence of the large lender due to a “relational overborrowing” effect. Borrowing terms with the large lender are determined by splitting the bilateral surplus. This surplus is largest when sovereign risk is present and the government pays high spreads on its market

¹As most of the quantitative literature on sovereign debt, we focus on debt as a vehicle for consumption smoothing or frontloading and abstract from project financing or development lending.

debt, precisely when the large lender is most needed. Consequently, when the large lender expects the government to face high spreads in the future, it values the relationship more and is willing to invest in it by lending more cheaply today. This endogenous elasticity of bilateral loan terms to indebtedness in markets acts as a countervailing force to the market discipline of spreads upon large issuances of debt. The presence of the large lender then leads the government to take on more debt in private markets and to delever more slowly. The end result is an increase in the default frequency and, therefore, in spreads, which ultimately create welfare losses for the government.

The dynamics and welfare effects we describe result from three critical assumptions regarding the large creditor and the funding it offers. Compared to competitive markets, we assume that it is more difficult (or impossible) to default on the large creditor, that its loans are of shorter maturity, and that the terms of borrowing result from bilateral bargaining rather than competition and a zero-profit condition. Central Bank swap lines constitute a prime real-world example, which [Horn et al. \(2021b\)](#) identify as a dominant channel through which official bilateral lending has reemerged since the 2000s.² These facilities are typically short-term and involve no recorded defaults (even concurrently with restructuring of market debts). Finally, while their terms are typically strictly confidential, anecdotal evidence suggests that a fair amount of bilateral negotiation is involved in determining amounts and interest rates, rather than market mechanisms.

Other forms of official debt share the seniority and/or duration features emphasized in our baseline model—including, for example, some types of IMF programs and repo or swap facilities offered by major central banks such as the Fed or the ECB. However, unlike the bilateral bargaining framework we assume, these facilities typically feature borrowing terms that are fixed in advance. For instance, IMF lending is subject to surcharges based on access and duration thresholds ([IMF, 2024b](#)), while the Fed’s swap lines carry a standard 25 basis point spread over policy rates ([Bahaj and Reis, 2023](#)) and are publicly reported in real time on the FRBNY’s [website](#). To accommodate such institutional designs, we develop an extension in which bilateral loan pricing does not result from bargaining but rather follows fixed rules. This allows us to explore how rules-based lending affects debt dynamics and to compare welfare outcomes across alternative institutional arrangements.

The extension shows, in a much simpler setup, that when the interest rate on bilateral loans is relatively insensitive to the government’s market debt—that is, when the interest rate is fixed or has a low elasticity to market debt—the relational overborrowing channel is muted or disappears altogether. In this case, the dominant effect of the large lender is to help the government

²While swap lines are important in our motivation to study the interaction of market debt and (undefaultable) bilateral loans, our model is not geared to represent all aspects of swap lines. For example, we do not include a difference in the currency denomination of bilateral and market debt, conditionality (including requirements on the level of foreign reserves), or collateral requirements.

avoid costly defaults on its marketable debt. As a result, the equilibrium features a lower default frequency and reduced sovereign spreads, improving the welfare of the borrowing country.

A key parameter in the model is the government's bargaining power relative to the large lender. When the government can make take-it-or-leave-it offers (i.e., when it has full bargaining power), the large lender simply provides nondefaultable loans at the risk-free rate, recovering the model of [Hatchondo, Martinez, and Önder \(2017\)](#). In this case, the government's welfare is also higher relative to the equilibrium without the large lender. In our preferred calibration, however, such gains quickly dissipate as the government's bargaining power declines.

In the baseline model, we assume that the large lender is purely motivated by profits. In fact, it shares the objectives, risk attitudes, and intertemporal preferences of the competitive creditors who lend in debt markets. We make this assumption not to represent any specific bilateral lender accurately, but rather to isolate the pure effect of market structure. However, the feedback from debt levels to the bilateral surplus, which gives rise to the relational overborrowing effect, will be present unless the large lender has a strong motivation to avoid default on the government's marketable debt. In this sense, the core results are robust to alternative descriptions of large bilateral creditors, including those with mixed policy, commercial, or financial stability objectives.

A key quantitative finding is that even relatively small bilateral loans can significantly affect sovereign debt outcomes when marketable debt has long maturity. The reason is that debt service on long-term debt is distributed over time, so the amount due in any given period is relatively small. By contrast, bilateral loans—modeled as short-term—must be repaid (or rolled over) every period. As a result, fluctuations in the bilateral loan interest rate have an impact on the current-period budget constraint that is comparable in magnitude to changes in the interest rate on a much larger stock of marketable debt.

The model delivers a simple heuristic for evaluating whether a particular form of bilateral lending is likely to be beneficial for the borrowing country. The key determinant is how the interest rate on bilateral loans responds to the level of the government's market debt: if bilateral loan terms improve as market debt rises, the relational overborrowing channel is likely to be active, as the large lender's willingness to lend cheaply amplifies the government's incentive to accumulate market debt. By contrast, if the terms of bilateral lending become more favorable when debt levels are low, it is more likely to support debt sustainability. These effects operate alongside more familiar channels related to avoiding improvements in the value of default, as recognized in policies like the IMF's Financing Assurances and Sovereign Arrears framework ([IMF, 2024a](#)), which aim to limit (bilateral) financing before at least a "credible process for debt restructuring is underway."

Discussion of the Literature We contribute to a nascent literature on the interaction of different types of sovereign debt.³ [Hatchondo, Martinez, and Önder \(2017\)](#) find that introducing a limited amount of non-defaultable debt improves the government’s welfare but only for a short period of time, while [Cordella and Powell \(2021\)](#) show how the preferred creditor status of international financial institutions can arise endogenously. [Dellas and Niepelt \(2016\)](#) focus on the coexistence of official and private creditors and show how an interior solution featuring types of lending can arise as a choice of the borrowing government. [Boz \(2011\)](#) and [Fink and Scholl \(2016\)](#) study the interaction of market debt with a type of senior debt coming with conditionality. [Kirsch and Rühmkorf \(2017\)](#) and [Roch and Uhlig \(2018\)](#) investigate the role of official lending or bailout agencies in eliminating equilibrium multiplicity, as do [Corsetti, Guimarães, and Roubini \(2006\)](#). The model we consider does not feature this type of equilibrium multiplicity, which would open the door to welfare gains of bilateral loans if they could help rule out unfavorable equilibria. Importantly, we also abstract from conditionality and focus on the impact of market design itself. Also related are [Kovrijnykh and Szentes \(2007\)](#) and [Faria-e-Castro et al. \(2024\)](#) who describe situations featuring strategic behavior of creditors in different contexts, and [Dovis and Kirpalani \(2023\)](#) who propose a design for a lender of last resort that improves the worst equilibrium.

More recently, [Arellano and Barreto \(2025\)](#) combine market and official debt in a model of partial default ([Arellano et al., 2023](#)) and find that competitive pricing and longer maturities, like those associated with bilateral lending from the Paris Club, endogenously make official debts less risky despite more favorable restructuring terms. [Liu, Liu, and Yue \(2025\)](#) show how the best subgame-perfect equilibrium of a sovereign debt model with market, bilateral, and multilateral creditors decentralizes the constrained-efficient allocation with imperfect information and moral hazard. We view these papers as emphasizing how certain institutions could potentially improve welfare in a world with both market and official debts, while our results exemplify the perils of bilateral sovereign debt. Also on the risks side, [Kondo, Sosa-Padilla, and Swaziek \(2025\)](#) study a sovereign debt model augmented by exogenous capital flows from a large lender (e.g. China) and focus on the risk that it demands repayment unexpectedly.

Empirically, [Perks et al. \(2021\)](#), and [Bahaj and Reis \(2021, 2023\)](#) document the network of Central Bank swap lines. [Cesa-Bianchi et al. \(2022\)](#) study swap lines among advanced economies, where they can serve a different purpose.

Layout The rest of the paper is structured as follows. Section 2 presents some motivating evidence on the increasing role of official external debt in emerging markets. Section 3 has two minimal models, highlighting the emergence of the cross-elasticity and how it can increase in-

³Excellent surveys of the broader literature on sustainable public debt and sovereign default can be found in handbook chapters by [Aguiar and Amador \(2014\)](#), [Aguiar, Chatterjee, Cole, and Stangebye \(2016\)](#), [D’Erasmus, Mendoza, and Zhang \(2016\)](#), and [Martinez, Roch, Roldán, and Zettelmeyer \(2023\)](#).

centives for debt dilution. Section 4 describes the main model in which both types of debt coexist, while Section 5 analyzes its equilibrium, and Section 6 describes the extension with fixed rules for the terms of bilateral loans. Finally, Section 7 concludes.

2. MOTIVATING EVIDENCE

Figure 1 summarizes the evolution of official external debt across emerging-market economies, using the World Bank’s International Debt Statistics (IDS). The data include all public and publicly guaranteed (PPG) external sovereign debt, measured in constant 2023 USD and aggregated across debtor countries. We distinguish between three broad official creditor categories: multilateral lenders (including the IMF and World Bank), Paris Club bilateral lenders,⁴ and other bilateral creditors—primarily non-Paris Club official lenders such as China.

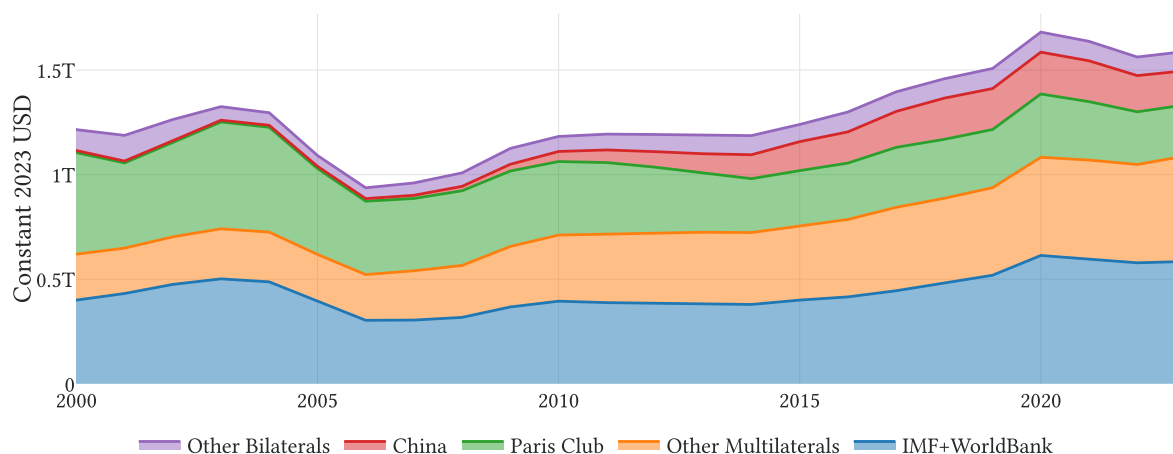


FIGURE 1: TOTAL OFFICIAL DEBT

Source: International Debt Statistics, The World Bank

The data show a clear shift in the composition of official external debt over the past two decades. While multilateral institutions and Paris Club members accounted for the bulk of official lending in the early 2000s, bilateral creditors outside the Paris Club have grown rapidly in both absolute and relative terms. By 2023, bilateral official lenders collectively account for roughly one-third of the global stock of official external debt to emerging markets, up from less than one-sixth two decades earlier. Among these, China has emerged as the single largest bilateral

⁴As of 2025, the members of the Paris Club are Australia, Austria, Belgium, Brazil, Canada, Denmark, Finland, France, Germany, Ireland, Israel, Italy, Japan, Netherlands, Norway, Russia, South Korea, Spain, Sweden, Switzerland, the United Kingdom, and the United States.

lender.

While the rise of new bilateral creditors is evident from Figure 1, it only reflects liabilities included within the official perimeters of PPG debt and hence reported to the World Bank. Arrangements with Central Banks (notably, swap lines) are often not included in such perimeters and therefore not captured in the figure.⁵ This limitation in the data illustrates the opacity of some of these deals (other aspects, like interest rates and maturities, are also typically confidential) which underpins concerns about the welfare implications of these new lenders (Horn et al., 2021a, 2023). Moreover, the scarce data on certain lending channels underscores the need for analytical tools that go beyond reported debt stocks to capture strategic interactions in sovereign borrowing decisions. The rest of our paper addresses these concerns through a theoretical and quantitative evaluation of a model that incorporates a large bilateral lender operating alongside competitive market creditors.

3. TWO MINIMAL MODELS

We begin our analysis by considering two stylized environments. These are designed to clarify the core mechanisms in the model described in Section 4. The first model describes how the cross-elasticity emerges as the government attempts to improve its bargaining position. The second model takes the cross-elasticity as given and characterizes how it magnifies the incentives for debt dilution, potentially leading to welfare losses.

3.1 A Two-Period Model: How the Cross-Elasticity Emerges from Bargaining

A key interaction in the quantitative model we introduce in Section 4 is the cross-elasticity of bilateral loan terms to market debt issuance. This elasticity arises as net debt issuances increase the government's cash on hand and enhance its threat point when bargaining with the senior lender. We illustrate this mechanism in a simplified setup by characterizing how the government's cash on hand influences its threat point and, through it, bargained borrowing terms.

There are two dates, $t = 0, 1$. The government receives endowments $y_0 < y_1$ and seeks to maximize an expected-utility objective with a strictly concave per-period utility function u and discount factor $\beta \in (0, 1)$:

$$V = u(c_0) + \beta u(c_1)$$

The senior (and only) lender is risk-neutral and discounts at $\beta_L \in (\beta, 1)$. All action occurs at the bargaining stage of period 0, when the government negotiates with the senior lender a

⁵Bahaj and Reis (2021, 2023) compile data on the network of Central Bank swap lines and in many cases are only able to see the total committed size of the swap lines, not drawn or borrowed amounts.

transfer of x in exchange for a repayment of $m' = x(1 + r)$ due in period 1. The government is able to commit to this repayment. With a Nash bargaining weight of θ for the lender, the bargaining problem is

$$\max_{x,r} \mathcal{L}(x,r)^\theta \times \mathcal{B}(x,r; y_0, y_1)^{1-\theta} \quad (1)$$

where

$$\begin{aligned} \mathcal{L}(x,r) &= -x + \beta_L x(1+r) = x(\beta_L(1+r) - 1) \\ \mathcal{B}(x,r; y_0, y_1) &= \underbrace{u(y_0 + x) + \beta u(y_1 - x(1+r))}_{\text{agreement}} - \underbrace{(u(y_0) + \beta u(y_1))}_{\text{threat point}} \end{aligned}$$

The focus of this section is to study the comparative statics of borrowing quantities x and terms r to the government's cash on hand y_0 . In the quantitative model, the government is able to affect its cash on hand by choosing its issuance of market debt. The strength of the cross-elasticity measures the incentive for the government to manipulate its threat point vis-à-vis the senior lender.

The solution to the bargaining problem (1) is characterized by two first-order conditions: a surplus-sharing rule and an Euler equation

$$\begin{aligned} u'(c_0) &= \frac{\theta}{1-\theta} \frac{\mathcal{B}(x,r; y_0, y_1)}{\mathcal{L}(x,r)} \\ u'(c_0) &= \frac{\beta}{\beta_L} u'(c_1) \end{aligned}$$

where $c_0 = y_0 + x$ and $c_1 = y_1 - x(1+r)$ are the on-path consumption levels.

The Euler equation means that the size of the transfer x should satisfy the borrower front-loading motive when using the lender's discount rate β_L^{-1} as the interest rate. In other words, the transfer is chosen to maximize total surplus. The surplus-sharing rule determines the terms at which the transfer is provided, setting the interest rate r to allocate surplus across lender and borrower.

Lemma 1. *Suppose the transfer size x is fixed at some $x^* > 0$. Then, higher cash on hand y_0 leads to better terms when bargaining with the senior lender: $\frac{\partial r}{\partial y_0} < 0$.*

Proof of Lemma 1 In this case the first-order condition is

$$\theta \beta_L \mathcal{B}(x^*, r; y_0, y_1) - (1 - \theta) \beta \mathcal{L}(x^*, r) u'(c_1) = 0$$

Omitting the arguments of $\mathcal{B}$ and $\mathcal{L}$ for readability, defining $F(r, y_0) = \theta \beta_L \mathcal{B} - (1 - \theta) \beta \mathcal{L} u'(c_1)$, and applying the implicit function theorem, we have

$$\frac{\partial r}{\partial y_0} = - \frac{\partial F / \partial y_0}{\partial F / \partial r} = - \frac{\theta \beta_L \frac{\partial \mathcal{B}}{\partial y_0}}{\theta \beta_L \frac{\partial \mathcal{B}}{\partial r} - (1 - \theta) \beta \frac{\partial \mathcal{L}}{\partial r} u'(c_1) + (1 - \theta) \beta \mathcal{L} u''(c_1) x}$$

For given (x, r) , strict concavity of u means that higher cash on hand y_0 reduces the government's surplus, so we have $\frac{\partial \mathcal{B}}{\partial y_0} < 0$, and the numerator is negative. For the denominator, higher interest rate r reduces the government's surplus, so $\frac{\partial \mathcal{B}}{\partial r} < 0$, and increases the lender's surplus, so $\frac{\partial \mathcal{L}}{\partial r} > 0$. Finally, the last term in the denominator is also negative, because of the concavity of u . In fact, the entire denominator being negative is the second-order condition for the maximization problem. We are then left with a negative numerator as well as negative denominator. with a minus sign in front, which yield a negative sign for the derivative $\frac{\partial r}{\partial y_0}$. $\square$

Lemma 2. *Suppose the interest rate r is fixed at some $r^* > 0$. Then, higher cash on hand y_0 leads to a lower reliance on the senior lender: $\frac{\partial x}{\partial y_0} < 0$.*

Proof of Lemma 2 In this case the first-order condition is

$$\theta \beta_L \mathcal{B}(x, r^*; y_0, y_1) - (1 - \theta) \beta \mathcal{L}(x, r^*) u'(c_1) = 0$$

Defining $F(x, y_0) = \theta \beta_L \mathcal{B} - (1 - \theta) \beta \mathcal{L} u'(c_1)$, and applying the implicit function theorem once more, we have

$$\frac{\partial x}{\partial y_0} = - \frac{\partial F / \partial y_0}{\partial F / \partial x} = - \frac{\theta \beta_L \frac{\partial \mathcal{B}}{\partial y_0}}{\theta \beta_L \frac{\partial \mathcal{B}}{\partial x} - (1 - \theta) \beta \frac{\partial \mathcal{L}}{\partial x} u'(c_1) + (1 - \theta) \beta \mathcal{L} u''(c_1)(1 + r)}$$

As in the previous case, we have $\frac{\partial \mathcal{B}}{\partial y_0} < 0$, and the denominator being negative is the second-order condition for the maximization problem, stating that the marginal benefit to the government of increasing the transfer is becoming lower than the marginal benefit to the lender as x increases above its optimal value. Once more, the relevant expression is the ratio of two negative numbers, with a minus sign in front. $\square$

In the general case when both x and r are free, an increase in y_0 continues to reduce the government's surplus, which leads to a fall in either x or r (or both) to restore the balance of the bargaining problem. In principle, the effect of reducing one of them could be so large to require an increase in the other variable. The conditions required to rule out these pathological cases seem to be quite weak and are satisfied for reasonable parametrizations of this problem (as well as in the quantitative model, as we confirm numerically in Section 5).

Discussion In the quantitative model of Section 4, in a given period, the government consumes its endowment plus the proceeds from market debt issuance and net transfers from the senior lender. At the beginning of the period, the government carries legacy debt b (which entails coupon payments of κb in the repayment state), owes m to the senior lender, and faces an income shock z . Higher legacy debt b maps into lower y_0 and y_1 and should in general make the government

weaker in the bargaining. However, higher revenues from debt issuance, map into higher y_0 and lower y_1 , both of which make the government stronger. The cross-elasticity arises when a higher choice of b' (on the left side of the debt Laffer curve) leads to more cash on hand in the current period and, thus, a stronger bargaining position.

3.2 A Three-Period Model: How the Cross-Elasticity Worsens Debt Dilution

We turn to analyzing the effect of the cross-elasticity on debt dilution. Debt dilution is a well-known feature of models with long-term debt, in which the issuer does not internalize losses

There are three dates, $t = 0, 1, 2$. In the final period, the government receives a stochastic endowment $z \sim N(0, \sigma_z)$, but wishes to consume in all periods to maximize an expected-utility objective

$$V = \mathbb{E} \left[\sum_{t=0}^2 u(c_t) \right] \quad (2)$$

for an increasing and concave utility function u satisfying Inada conditions.

To finance consumption in all three periods, the government issues long-term bonds b at $t = 0$ as well as short-term bonds d and a bilateral loan m at $t = 1$. All three are expressed as promises to repay one unit of the good at $t = 2$. While the loan m is undefaultable, the government faces a choice of whether to repay the bonds, subject to some costs $h(z)$. In other words, consumption in period 2 is given by

$$c_2 = \begin{cases} y(z) - m - (b + d) & \text{if repay} \\ y(z) - m - h(z) & \text{if default} \end{cases}$$

In this finite-horizon example, since the utility function is increasing, default occurs exactly in those states for which the costs of default fall short of the value of the debt. In other words,

$$c_2 = y(z) - m - \min\{b + d, h(z)\} \quad (3)$$

In the middle period, the government issues short-term debt d and loans m . Thanks to competition among risk-neutral investors, the price of debt reflects the probability of default so $q(b, d) = \mathbb{P}(b + d < h(z)) = q(b + d)$ with a slight abuse of notation. Notice that the price of short-term debt depends also on the outstanding amounts of long-term debt (in a perfectly-substitutable way as default affects both types of bond equally). We assume an exogenous rule for the pricing of bilateral loans, under which more legacy debt b may induce a lower price but more recently-issued debt d induces higher prices (the cross-elasticity), exactly as suggested by the bargaining model of Section 3.1. To keep things simple, we assume

$$\phi(b, d, m) = \alpha_0 + \alpha_b b + \alpha_d d + \alpha_m m$$

where $\alpha_d > 0$, as well as a fixed debt limit $\bar{m}$. Consumption in the middle period is

$$c_1 = dq(b + d) + m\phi(b, d, m) \quad (4)$$

Finally, in the initial period, the government issues long-term debt b at price q , but investors hold beliefs d^e about the issuance of short-term bonds d which will happen later. Initial consumption is

$$c_0 = bq(b + d^e) \quad (5)$$

for the same price function q as above (notice that, since the bilateral loan m does not affect the default set, there is no need to anticipate m to decide q).

We consider different cases for the government's ability to influence beliefs d^e .

Commitment When the government can commit, it chooses $\{b, m, d^e, d\}$ to maximize (2) subject to (3), (4), (5), $m \leq \bar{m}$ and $d = d^e$.

The first-order conditions for m (using λ to denote the multiplier on the debt limit constraint) and b are

$$\begin{aligned} u'(c_1) \left(\phi + m \frac{\partial \phi}{\partial m} \right) &= \mathbb{E} [u'(c_2)] + \lambda \quad (6) \\ u'(c_0) \left(bp'(b + d) + p(b + d) \right) + u'(c_1) \left(dp'(b + d) + m \frac{\partial \phi}{\partial b} \right) &= q(b + d) \mathbb{E} [u'(c_2^R)] \end{aligned}$$

and the first-order condition for d is

$$u'(c_0)bp'(b + d) + u'(c_1) \left(dp'(b + d) + p(b + d) + m \frac{\partial \phi}{\partial d} \right) = q(b + d) \mathbb{E} [u'(c_2^R)] \quad (7)$$

where we highlight that, because of equal seniority of short and long term debt, the contribution of period 2 to the first-order conditions for b and d is the same.

Discretion When deciding sequentially, the government takes as given a value d^e and chooses $\{b, m, d\}$ to maximize (2) subject to (3), (4), (5) and $m \leq \bar{m}$. Equilibrium then requires investors to be correct in their assessment, so that the government does choose $d = d^e$. As a result, the first-order conditions for both b and m remain unchanged, but the first-order condition for d is now 'missing' the terms reflecting the manipulation of d^e by the committed government:

$$u'(c_1) \left(dq'(b + d) + q(b + d) + m \frac{\partial \phi}{\partial d} \right) = q(b + d) \mathbb{E} [u'(c_2^R)] \quad (8)$$

How the cross-elasticity worsens debt dilution With commitment and $\bar{m} = 0$, our assumptions (no discounting and the fact that the technology for defaulting both types of bonds b and d is the same) mean that the government would like to set $b = d$ and $c_0 = c_1$: combining the first-order conditions for b and d in the case of commitment when $\bar{m} = 0$ yields

$$u'(\underbrace{q(b+d)b}_{=c_0}) (bq'(b+d) + q(b+d) - q'(b+d)) = u'(\underbrace{q(b+d)d}_{=c_1}) (dq'(b+d) + q(b+d) - q'(b+d))$$

which highlights the symmetry between both periods and both types of bond.

In the case of discretion, the missing time-0 term in (8) induces a desired increase in d at the expense of b , as increasing d increases the default probability and worsens debt prices, which induces lower b .⁶ At $\bar{m} = 0$, the difference between d^D and d^C measures the extent of debt dilution coming only from the presence of long-term debt.

As $\bar{m}$ increases, if it binds and $m^D = \bar{m}$,⁷ we would have c_1 increasing in $\bar{m}$. This means that, when comparing (7) with (8), the removal of a cost of short-term debt d remains the same, but the term which is used to compensate is now multiplied by a smaller marginal utility $u'(c_1)$, so a larger compensation is required. This means that the incentive to issue more short-term debt (from the commitment solution) is increasing in $\bar{m}$.

After increasing d , the government would observe a higher default probability, so it would like to respond by decreasing b , which brings a stabilizing effect on prices. If $\alpha_b < 0$, like we obtain in the quantitative version (because more legacy debt makes the government weaker in the bargaining), moving $\bar{m}$ will also affect the commitment solution to begin with. In that case the government would already start to decrease b and increase d as $\bar{m}$ increases.

Figure 2 illustrates how the cross-elasticity worsens dilution. It shows the solutions to the problems of commitment and discretion (as well as the choices of the sequential government if it faced the prices from the problem with commitment) as a function of a maximum amount $\bar{m}$ that can be drawn on the bilateral loan.

For every value of $\bar{m}$, as anticipated in our discussion the government chooses a higher debt issuance d_{DC} in the middle period when it reoptimizes but continues to face the prices from the solution with commitment, relative to the commitment solution d_C . The gap between d_C and d_{DC} is also increasing with $\bar{m}$, owing to the fact that the partial-equilibrium incentives in (7) are an increasing function of m . The Figure also shows that in this example, the effect of prices induces further increases in the amount issued under discretion d_D relative to d_{DC} . These forces combine to boost the losses from debt dilution: the gap between the value attained under discretion and commitment, shown in the left panel of Figure 2, clearly increases with $\bar{m}$ while the constraint $m \leq \bar{m}$ binds.

⁶This is stabilizing effect as lower b improves prices marginally which may increase d a bit again.

⁷If the limit on m does not bind, from (6), a larger d in the case of discretion will push ϕ up, inducing $m^D > m^C$.

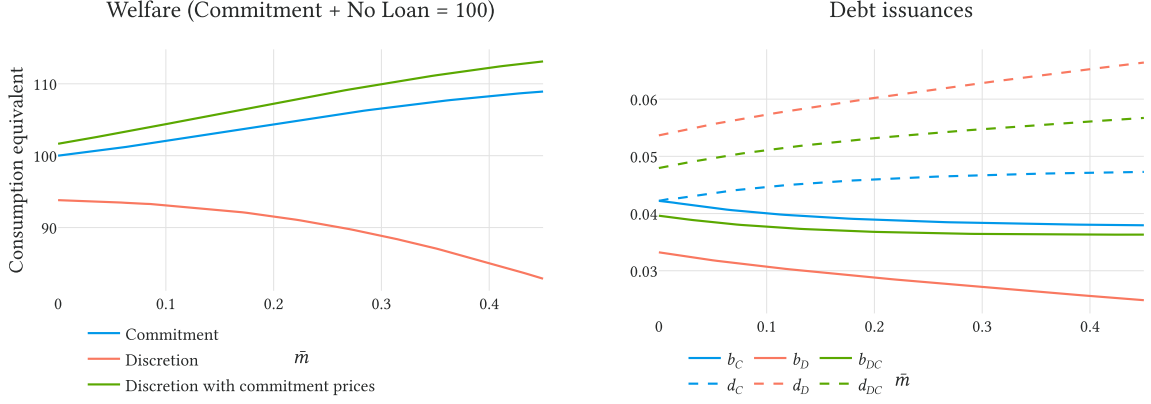


FIGURE 2: DEBT DILUTION AND THE CROSS-ELASTICITY

4. MODEL WITH BILATERAL LOANS AND DEFAULTABLE MARKET DEBT

In this section we present the full version of the model, in which the borrowing government has access to the large lender as well as to a competitive fringe of lenders. Default on the debt b held by competitive lenders is possible, subject to standard output costs of default. However, just as before, bilateral loans m cannot be defaulted.

The timeline of events within a period are illustrated in Figure 3.

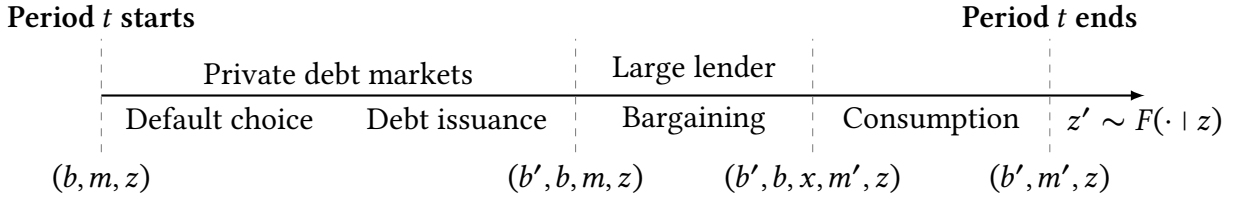


FIGURE 3: TIMELINE OF EVENTS WHILE NOT IN DEFAULT

At the start of t , the government owes m to the large lender, b to the market, and observes the exogenous state z . Additionally, the economy can be in default ($\zeta = D$) or in repayment ($\zeta = R$). Let $v(b, m, z)$ and $h(b, m, z)$ represent the government's and the large lender's value functions at the beginning of a period, when the government is in good standing with the market.

Private markets In the morning of t , first, the government decides default for the current period if it is in good standing with the market.

$$v(b, m, z) = \max \{v_R(b, m, z) + \epsilon_R, v_D(m, z) + \epsilon_D\}, \quad (9)$$

where the ϵ 's follow a Type 1 Extreme Value distribution, yielding closed forms for $v(b, m, z)$ and the (beginning-of-period) default probability $P(b, m, z)$

$$v(b, m, z) = \chi \log \left(\exp(v_D(m, z)/\chi) + \exp(v_R(b, m, z)/\chi) \right)$$

$$P(b, m, z) = \frac{\exp(v_D(m, z)/\chi)}{\exp(v_D(m, z)/\chi) + \exp(v_R(b, m, z)/\chi)}$$

We assume that defaulting on the market debt fully destroys all bonds outstanding and triggers exclusion from international capital markets. Access to markets is then recovered stochastically with constant probability ψ . In addition, the maximum amount that can be borrowed from the large lender m' while in default on its market debt is constrained by $m' \leq \Gamma(m, z)$. We begin our analysis without restrictions by setting $\Gamma(m, z) = +\infty$ but describe below different possibilities for the value of Γ .

If default is not chosen in the current period, the government issues new debt b' to the fringe of lenders understanding the value of entering negotiations with the monopolist having issued this debt level b' ,

$$v_R(b, m, z) = \max_{b'} w_R(b', b, m, z), \quad (10)$$

where debt is a perpetuity of geometrically-decaying coupons (Leland, 1998; Hatchondo and Martinez, 2009; Arellano and Ramanarayanan, 2012). A unit of debt issued in period t promises to repay $\kappa(1 - \delta)^{s-1}$ units of the tradable good in period $t + s$, effectively making a unit issued at $t - 1$ a perfect substitute for $(1 - \delta)$ units issued at t . In this sense, δ controls the maturity of debt while κ controls the average size of the coupon payments.

The price faced by the borrower government reflects its lenders' expectations of repayment, discounted with a risk-neutral kernel

$$q(b', b, m, z) = \frac{1}{1 + r^*} \mathbb{E} [(1 - \mathbb{1}_D(b', m', z')) (\kappa + (1 - \delta)q(b'', b', m', z')) | z] \quad (11)$$

where r^* is the international risk-free rate,⁸ $\mathbb{1}_D(b, m, z)$ denotes the government's default policy as perceived by the competitive lenders, $m' = m'_R(b', b, m, z)$ is the expected result of negotiations with the large lender, to happen in the 'afternoon' of period t , and $b'' = g_b(b', m'_R(b', b, m, z), z')$ is the expected debt issuance in the following period.

Bilateral loan In the afternoon of t , the government meets with the large lender to negotiate the bilateral loan. As before, the outcome of their negotiation is a transfer x and new loan size m' which solve the following Nash bargaining problems, depending on default status vis-à-vis the

⁸We set $\kappa = r^* + \delta$, so that the price of debt equals 1 when the government repays with certainty.

market

$$\begin{aligned} & \max_{m', x} \mathcal{L}_R(b', x, m, m', z)^\theta \mathcal{B}_R(b', b, x, m, m', z)^{1-\theta} \\ & \text{or} \\ & \max_{m', x} \mathcal{L}_D(x, m, m', z)^\theta \mathcal{B}_D(x, m, m', z)^{1-\theta} \end{aligned} \quad (12)$$

As before, the large lender's surplus is

$$\begin{aligned} \mathcal{L}_R(b', x, m, m', z) &= -x - m + \beta_L \mathbb{E} [h(b', m', z') - h(b', 0, z') \mid z] \\ \mathcal{L}_D(x, m, m', z) &= -x - m + \beta_L \mathbb{E} [\psi (h(0, m', z') - h(0, 0, z')) + (1 - \psi) (h_D(m', z') - h_D(0, z')) \mid z] , \end{aligned}$$

while the borrower's surplus now also reflects outcomes in debt markets

$$\begin{aligned} \mathcal{B}_R(b', b, x, m, m', z) &= u(y(z) + B(b', b, m, z) + x) - u(y(z) + B(b', b, m, z) - m) + \\ & \quad + \beta \mathbb{E} [v(b', m', z') - v(b', 0, z') \mid z] \\ \mathcal{B}_D(x, m, m', z) &= u(y_D(z) + x) - u(y_D(z) - m) + \\ & \quad + \beta \mathbb{E} [\psi (v(0, m', z') - v(0, 0, z')) + (1 - \psi) (v_D(m', z') - v_D(0, z')) \mid z] \end{aligned}$$

where the function $y_D(z) = y(z) - \xi(z)$ is output in default and $B(b', b, m, z)$ summarizes net transfers from the competitive lenders received in the morning. Our assumptions on debt maturity yield $B(b', b, m, z) \equiv q(b', b, m, z)(b' - (1 - \delta)b) - \kappa b$. In default, opportunities to reaccess markets arrive with probability ψ . The bargaining problems yield new terms for the bilateral loan $\{x_R(b', b, m, z), m'_R(b', b, m, z)\}$ and $\{x_D(m, z), m'_D(m, z)\}$ in repayment and in default, respectively.

The most important way in which the presence of debt markets affects the bargaining stage is through the net flows from debt repayment and new issuance, $B(b', b, m, z)$. These flows modulate the government's threat point: after a successful issuance which raises large revenues, the government is in a strong position to negotiate as repaying m is less costly.

Consumption After the negotiation is done and transfers settled, consumption takes place.

$$\begin{aligned} c_R(b', b, m, z) &= y(z) + B(b', b, m, z) + x_R(b', b, m, z) \\ c_D(m, z) &= y(z) - \xi(z) + x_D(m, z) \end{aligned} \quad (13)$$

Given transfers $x_R(b', b, m, z)$ and $x_D(m, z)$, we can now return to the beginning of the period and determine the values of repaying and defaulting

$$\begin{aligned} w_R(b', b, m, z) &= u(c_R(b', b, m, z)) + \beta \mathbb{E} [v(b', m'_R(b', b, m, z), z') \mid z] \quad \text{and} \\ v_D(m, z) &= u(c_D(m, z)) + \beta \mathbb{E} [\psi v(0, m'_D(m, z), z') + (1 - \psi) v_D(m'_D(m, z), z') \mid z] , \end{aligned} \quad (14)$$

while for the large lender we have

$$\begin{aligned}
h(b, m, z) &= P(b, m, z)h_D(m, z) + (1 - P(b, m, z))h_R(b'(b, m, z), b, m, z), \\
h_R(b', b, m, z) &= a - x_R(b', b, m, z) + \beta_L \mathbb{E} [h(b', m'_R(b', b, m, z), z') \mid z] , \text{ and} \\
h_D(m, z) &= a - x_D(m, z) + \beta_L \mathbb{E} [\psi h(0, m'_D(m, z), z') + (1 - \psi)h_D(m'_D(m, z), z') \mid z] .
\end{aligned} \tag{15}$$

5. QUANTITATIVE RESULTS

We parametrize our model at a quarterly frequency following standard strategies in the sovereign default literature. Most parameters are set externally, such as those governing the stochastic process for income, risk-free rates, risk aversion, the length of exclusion after default, and the duration of market debt (Chatterjee and Eyigungor, 2012).⁹

The critical parameter θ , which governs the lender's bargaining power, is set to 0.5 in our baseline but we study comparative statics around it later on. The remaining parameters (the sovereign's discount rate and the default costs) are calibrated internally to match the average debt-to-output ratio as well as the average and standard deviation of spreads. Our calibration is based on comparing data for Argentina between 1993q1 and 2001q4 to simulated data from the model. For the simulations, we use 35-period samples that end just before a default event, and condition on no default occurring within those 35 periods or two years prior, as is standard in this literature when assuming no recovery after default (Arellano, 2008; Chatterjee and Eyigungor, 2012; Roch and Roldán, 2023).

Table 1 summarizes our parametrization.

Table 2 presents some statistics from simulating the model with and without bilateral loans, for different values of the lender's bargaining weight θ . Statistics correspond to pre-default samples of 35 quarters and the term 'loans' refers to bilateral loans m . Welfare gains are reported in equivalent consumption terms and calculated averaging the ergodic distribution of the equilibrium in which only market debt is allowed, conditional on repayment. These welfare gains arise from comparing the economy with only market debt with economies with both types of debt at $m = 0$. The leftmost column corresponds to the equilibrium with only market debt and as such generates the business-cycle properties of Argentina in the years preceding its 2001 default.

The remaining columns show that even with a relatively large bargaining weight for the government, the availability of bilateral loans substantially increases the frequency of default. As a consequence, the government pays higher and more volatile spreads, despite slightly lower market debt levels and relatively modest amounts borrowed from the large lender. As a result,

⁹As has also become standard for numerical performance, we include preference shocks for default as described above, but also for the debt issuance choice; both are set to a small variance to avoid impacting the equilibrium.

TABLE 1: BASELINE PARAMETER VALUES

	Parameter	Value
Sovereign's discount factor	β	0.9504
Sovereign's risk aversion	γ	2
Preference shock scale parameter	χ	0.025
Lender's bargaining power	θ	0.5
Risk-free interest rate	r^*	0.01
Duration of debt	δ	0.05
Income autocorrelation coefficient	ρ_z	0.9484
Standard deviation of y_t	σ_z	0.02
Reentry probability	ψ	0.0385
Default cost: linear	d_0	-0.24
Default cost: quadratic	d_1	0.3

TABLE 2: BUSINESS CYCLE STATISTICS WITH AND WITHOUT BILATERAL LOANS

	Only market	Unrestricted, $\theta = 0.25$	Unrestricted, $\theta = 0.5$
Avg spread (bps)	714	1,613	2,105
Std spread (bps)	399	927	1,331
$\sigma(c)/\sigma(y)$ (%)	113	109	109
Debt to GDP (%)	22.5	21.7	21.2
Loan to GDP (%)	0	3.4	3.02
Loan spread (bps)	–	-52.5	-429
Corr. loan & spreads (%)	–	61.7	67.5
Default frequency (%)	5.72	11	13
Welfare gains (rep)	–	-0.15%	-0.43%

Note: Statistics are based on model simulations of 35 quarters before a default on market debt. The column marked ‘Only market’ refers to a version of the model without bilateral loans, while the columns marked ‘Unrestricted’ refer to versions in which bilateral loans can be used even during default episodes. Default frequencies are computed on the ergodic distribution. Welfare gains are given as equivalent increases in consumption calculated over the ergodic distribution of the model without bilateral loans, conditional on repayment.

even with $\theta = 0.25$, the government prefers the equilibrium in which the large lender is not present.

Since the large lender keeps a share of the surplus generated by the loan, the borrower government is somewhat reluctant to use it. In a typical simulation path, conditional on no default the amount borrowed bilaterally is 3.3% of annual income with a standard deviation of 1.6%. Figure 4 shows that this changes significantly around default events: the loan size m shoots up around the moment of default. The large lender heavily subsidizes this accumulation of bilateral debt. As before, the large lender provides negative interest rates at first, while bilateral debt is increas-

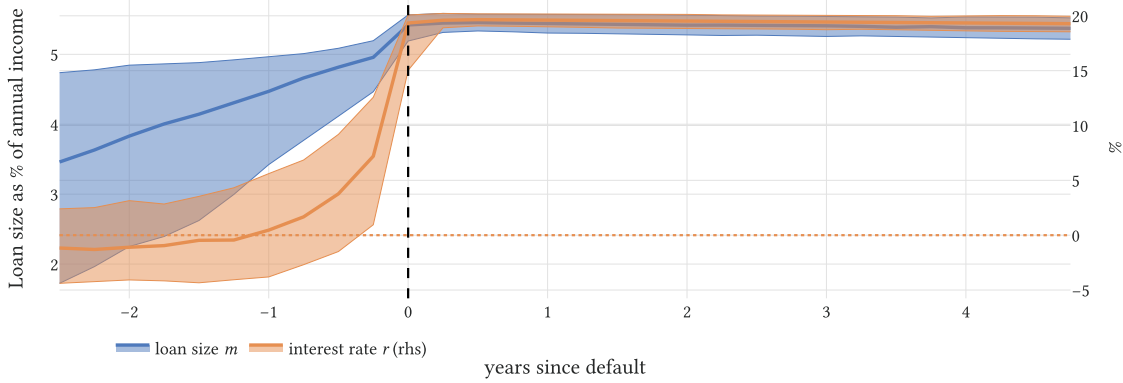


FIGURE 4: LOANS AROUND DEFAULT EVENTS

ing, to then raise them as default grows closer. In this case, the gamble for debt overhang has a twist: when the economy recovers market access, it immediately issues market debt to pay off the bilateral loan (see Figure 12 in the Appendix). The large lender is then gambling that income will not revert and that the exclusion period will be long.

While most of the use of the bilateral loan m occurs during default, Figure 4 shows that default episodes are preceded by bilateral issuance in an effort to avoid or postpone default (Figure 4 does not show the defaults that were avoided as a consequence of the large lender's presence).

Bilateral loans affect the economy in two ways: on the one hand, they provide extra financing when default risk makes borrowing in private markets costly. But they also provide funds in default, which raises the value of being excluded from private markets and consequently spreads. To disentangle these effects, we consider a variant of the model in which the large creditor is only willing to lend while the economy is in good standing with private debt markets, in other words, a variant where we set $\Gamma(m, z) = 0$. This implies in particular that whenever the government defaults, it must at the same time repay all of its bilateral loans.

Table 3 presents some statistics from simulating the model with and without bilateral loans (now marked 'Unrestricted'), as well as the variant in which bilateral loans are 'Limited.'

In the version with Limited bilateral loans in default, the usage of these loans declines by more than two-thirds on average, as repaying the large creditor after defaulting becomes even

TABLE 3: BUSINESS CYCLE STATISTICS WITH BILATERAL LOANS

	Only market	Unrestricted, $\theta = 0.5$	Limited, $\theta = 0.5$
Avg spread (bps)	714	2,105	1,038
Std spread (bps)	399	1,331	612
$\sigma(c)/\sigma(y)$ (%)	113	109	113
Debt to GDP (%)	22.5	21.2	22.5
Loan to GDP (%)	0	3.02	1.06
Loan spread (bps)	–	-429	536
Corr. loan & spreads (%)	–	67.5	71.1
Default frequency (%)	5.72	13	7.72
Welfare gains (rep)	–	-0.43%	-0.2%

Note: Statistics are based on model simulations of 35 quarters before a default on market debt. ‘Only market’ refers to a version of the model without bilateral loans, ‘Unrestricted’ to a version in which bilateral loans can be used even during default episodes, and ‘Limited’ to a version in which the bilateral loan must have a balance of 0 during default. Default frequencies are computed on the ergodic distribution, as are welfare gains, which are given as equivalent increases in consumption calculated over the ergodic distribution of the model without bilateral loans, conditional on repayment.

more costly (and can even act as an extra cost of default). However, bilateral loans are still mostly used when spreads are high, as shown by the high correlation between spreads and loans. The Limited version still creates welfare losses for the economy, but these are much lower than in the Unrestricted case.

5.1 Default probabilities and debt prices

Figure 5 shows ex-post default regions for private debt, when the bilateral loan $m = 0$. Solid lines represent the case without bilateral loans, while dotted and dashed lines correspond to the versions with Unrestricted and Limited bilateral loans. When bilateral loans are Unrestricted, their presence leads to an increase in default as the government is able to sustain lower levels of debt. The Limited loans case has government default policies which are virtually unchanged from the case without bilateral loans.

Figure 5 highlights the negative impact of the bilateral loans. In the Unrestricted version, loans are available during default, which makes market debt repayment less attractive. The higher default probability translates into lower prices for debt, as shown in Figure 6. This effect is muted

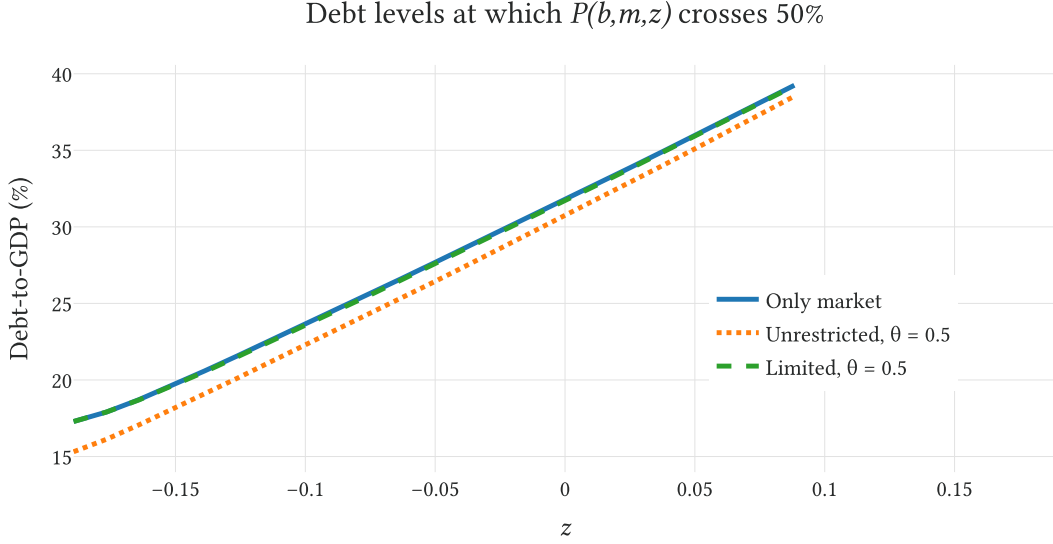


FIGURE 5: DEFAULT REGIONS

in the Limited variant, when bilateral loans are not available in default. However, as shown in Figure 6, prices remain lower in the Limited variant relative to the model without the large creditor, especially when debt is low. This means that even though the one-period-ahead default probability may not increase when bilateral loans are introduced in a limited fashion, policies (notably, future borrowing) are altered in a way that still creates more default risk later on. In other words, the option of bilateral loans, even in the Limited variant, increases debt dilution and therefore lowers equilibrium debt prices.

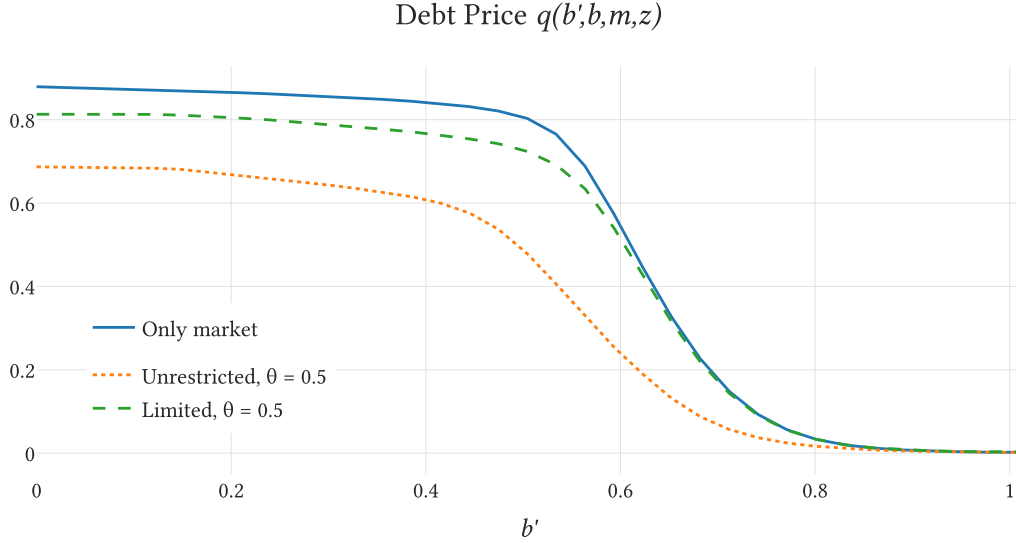


FIGURE 6: DEBT PRICES

5.2 Dynamics with bilateral loans

In the model with bilateral loans, the government issues debt in a riskier manner. Figure 7 presents the ergodic distribution of the debt-to-GDP ratio in simulations of the three models, conditional on repayment. When bilateral loans are available, in both variants, the economy spends more time in the region of the state space where default risk is large.

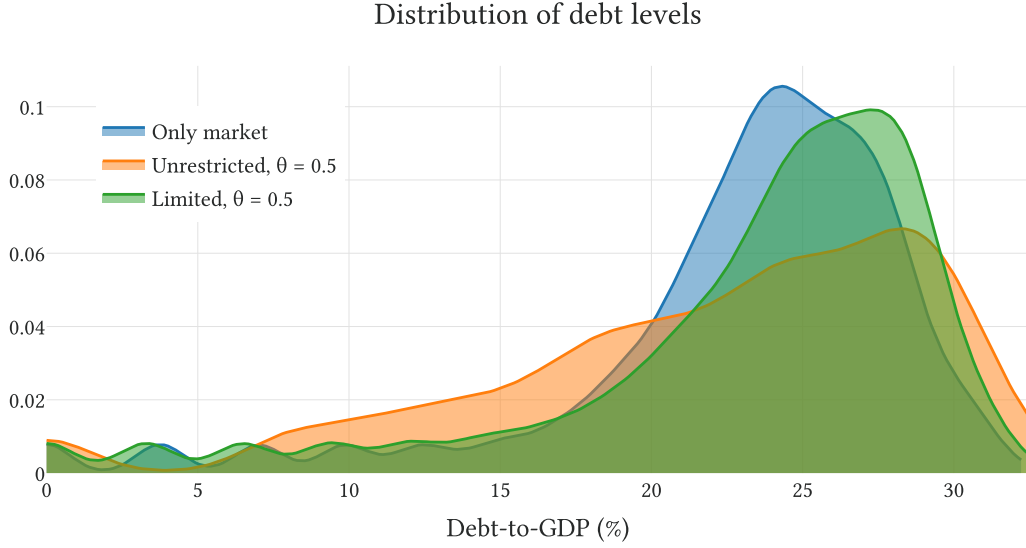


FIGURE 7: DISTRIBUTION OF DEBT LEVELS

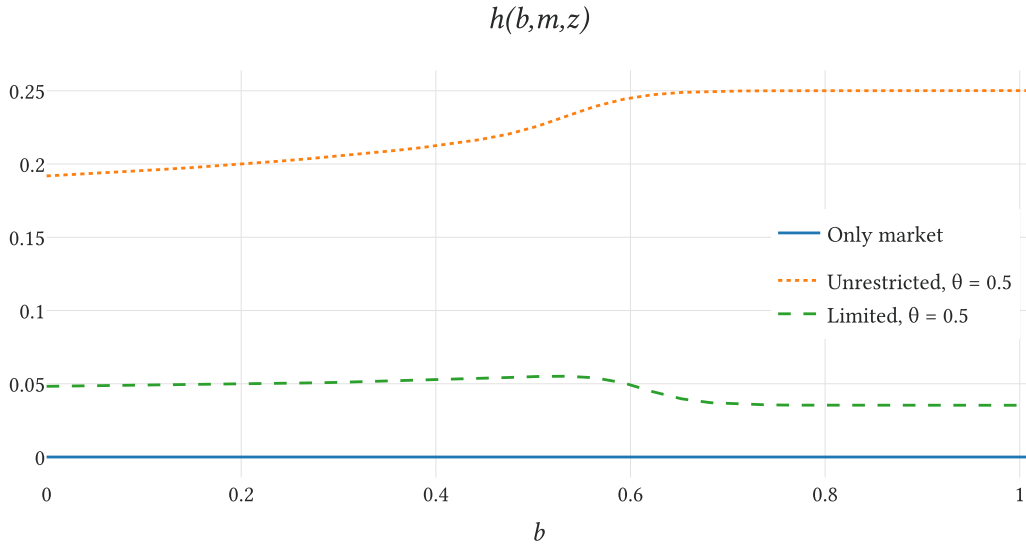


FIGURE 8: LARGE LENDER'S PROFITS

Figure 8 plots the monopolist's value (or profit) function (15) as a function of debt, keeping $m = 0$ and the endowment level at its mean. In both versions (both when loans are Unrestricted

or Limited in default), the large lender's profits are increasing in market debt b , at least when debt is low enough that the one-period-ahead default probability remains contained. In this region, as spreads open up, the presence of the large lender is more valuable to the government, which increases total surplus in the negotiations. As debt continues to grow and default becomes more likely, profits for the large lender decrease in the Limited version, as any bilateral lending m must be repaid at face value upon default, but increase sharply in the Unrestricted case, in which large payments can be extracted from the borrower during the default spell.

When the government has a safe level of debt and pays low spreads, there is little for the monopolist to offer that private markets do not already provide at a competitive price. But when default risk drives up spreads, the monopolist can provide financing through the bilateral loan, which is non-defaultable, and charge an interest rate between the risk-free rate and the rate that the government is paying on its market debt.

Relational overborrowing At the bargaining stage, the choice of market debt b' has already been made by the government. As a result, if bilateral negotiations break down the borrower is left with a consumption level of $y(z) + q(b', b, m, z)(b' - (1 - \delta)b) - \kappa b - m$. In periods when the government is trying to reduce market debt b' , it enters negotiations with the large lender in a 'weak' position as consumption at the government's threat point is low. This generates large gains (for the borrower) from any transfer x provided by the large lender. Thus, in order to split the surplus such transfers must come at a high interest rate. Conversely, when the government has issued a large amount of bonds b' , it expects a high level of consumption even if negotiations break down, which results in a strong position and good borrowing terms with the large lender.

Figure 9 shows the interest rate that results from bilateral negotiations as a function of the choice of b' . It confirms the discussion above: borrowing terms with the large lender respond aggressively to the government's choice of indebtedness in debt markets.

As a result of this elasticity of loan terms to debt, new terms appear in the government's Euler equation for market debt b' :

$$u'(c) \left(q + \frac{\partial q}{\partial b'} i + \frac{\partial x}{\partial b'} \right) = \beta \mathbb{E} \left[u'(c')(1 - \mathbb{1}_D) \left(\kappa + (1 - \delta)q' - i' \left(\frac{\partial q'}{\partial b} - \frac{\partial q'}{\partial m} \frac{\partial m'}{\partial b'} \right) - \frac{\partial x'}{\partial b} + \frac{\partial x'}{\partial m} \frac{\partial m'}{\partial b'} \right) \right] \quad (16)$$

where $i = b' - (1 - \delta)b$ represents new market debt issuances, $\frac{\partial q'}{\partial b}$ refers to the derivative of the following period's debt price with respect to the beginning-of-period debt level at that point (in other words, b' from the perspective of the current period) and similarly for $\frac{\partial q'}{\partial m}$, $\frac{\partial x'}{\partial b}$, and $\frac{\partial x'}{\partial m}$. Relative to a world with only market debt, there are three new effects arising from the presence of the large lender.

First, issuing debt affects transfers x received from the large lender in the afternoon of the

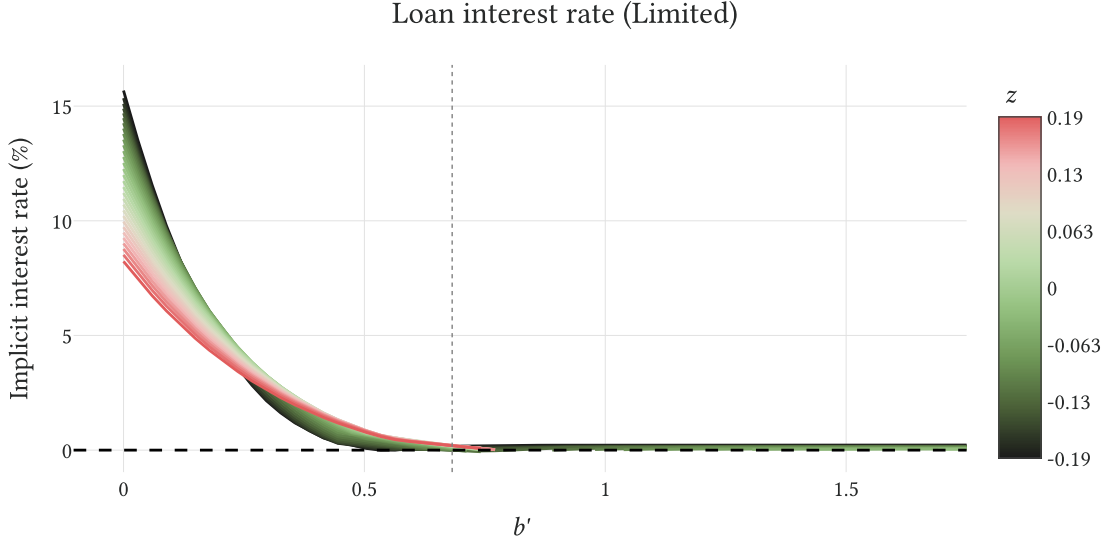


FIGURE 9: INTEREST RATE ON THE BILATERAL LOAN

current period. Second, issuing debt also affects the desired amount of loans m' taken from the large creditor. This flows into debt prices q' and transfers x' in the following period through the chain-rule terms. Finally, because debt prices and borrowing terms also depend on the initial level of indebtedness b , issuing new debt affects both the price at which market debt will be placed q' as well as transfers x' in the following period.

The left-hand side of (16) describes a countervailing force to the market discipline of spreads. In a model with only market debt, the benefit of issuing an extra unit of debt is $u'(c) \left(q + \frac{\partial q}{\partial b'} i \right)$: the government understands that issuing more debt may increase spreads (decrease the price q) and this mitigates issuances. With the large lender, the benefit of issuing an extra bond includes additionally the term $\frac{\partial x}{\partial b'}$ as the extra issuance also affects the threat point at the bargaining stage. Notice that, by writing the transfer x as a price $\frac{1}{1+r}$ times the new loan m' (as in (18)), the marginal benefit of debt term becomes $u'(c) \left(q + \frac{\partial q}{\partial b'} i + \frac{1}{1+r} \frac{\partial m'}{\partial b'} + \frac{\partial \frac{1}{1+r}}{\partial b'} m' \right)$, which emphasizes how debt issuances affect revenues from the market and from the large lender in a strictly symmetrical way (notice also that $\frac{\partial i}{\partial b'} = 1$).

5.3 Welfare effects of bilateral loans

The forces discussed above combine to produce the welfare effects of bilateral loans. Figure 10 shows the government's value function as a function of debt b when it owes $m = 0$ to the large lender. The government prefers bilateral loans to be Unavailable during default, except of course when a default in the current period is very likely.

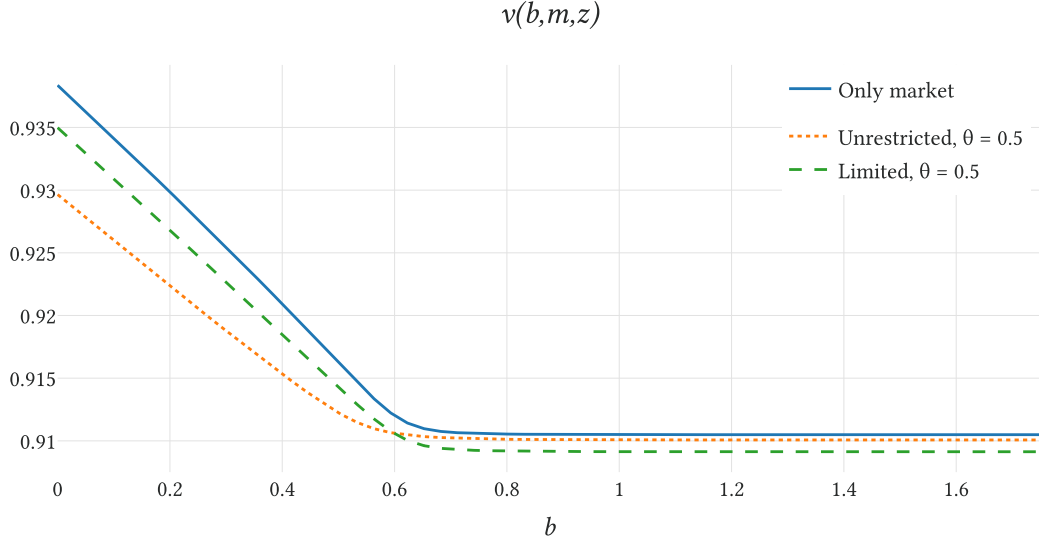


FIGURE 10: VALUE FUNCTIONS

6. PROGRAMMING THE LARGE LENDER

We have shown how the combination of market debt and bilateral loans can create welfare losses for the government through the relational overborrowing effect which incentivizes risk-taking in debt issuances.

In this section we replace the bargaining protocol with fixed rules for the terms of bilateral loans, with two objectives. First, to better understand the dynamics of the relational overborrowing effect in a simpler case. Second, to explore design questions surrounding bilateral loans. Can rules that contain spreads on market debt by curbing sovereign default risk be found? Are there designs that can create welfare gains for the government? What is the optimal design of bilateral loans? Are there possible Pareto improvements, under which the government attains a higher value than without bilateral loans and also the large lender obtains profits?

To investigate these questions, we remove the bargaining protocol for the bilateral loans and, instead, consider rules of the form

$$r(b', m') = \max\{r^*, \alpha_0 + \alpha_b b' + \alpha_m m'\}$$

With these rules, the government now faces the exogenous interest rate $r(b', m')$ (on the bilateral debt) when it chooses market debt b' and bilateral loan m' . Notice that because the bilateral loan always costs weakly more than the risk-free rate r^* , the large lender makes non-negative profits.

At this moment, we consider a ‘risk-inducing’ rule with $\alpha_0 > 0$, $\alpha_b < 0$, $\alpha_m = 0$, which replicates the main properties of the equilibrium with bargaining, and a size-dependent rule which does not

load on indebtedness in markets, with $\alpha_0 > 0, \alpha_b = 0, \alpha_m > 0$. Table 4 summarizes our findings.

TABLE 4: OUTCOMES WITH EXOGENOUS RULES FOR BILATERAL LOANS

	Only market	Size dependent r	Risk inducing r	Limited, $\theta = 0.5$
Avg spread (bps)	714	623	921	1,038
Std spread (bps)	399	315	552	612
$\sigma(c)/\sigma(y)$ (%)	113	115	115	113
Debt to GDP (%)	22.5	23.5	22.8	22.5
Loan to GDP (%)	0	0.71	0.972	1.06
Loan spread (bps)	–	682	1,264	536
Corr. loan & spreads (%)	–	62.5	48.1	71.1
Default frequency (%)	5.72	5.13	6.92	7.72
Welfare gains (rep)	–	0.21%	-0.079%	-0.2%

Note: Statistics are based on model simulations of 35 quarters before a default on market debt. ‘Only market’ refers to a version of the model without bilateral loans, while ‘Limited’ refers to the version in which bilateral loan balances must be 0 while in default. The middle columns denote the versions with exogenous terms for the bilateral loan: ‘Size dependent r ’ has the bilateral interest rate increase with the balance in the bilateral loan, while ‘Risk inducing r ’ has the bilateral interest rate decrease with the amount of market debt. Default frequencies are computed on the ergodic distribution. Welfare gains are given as equivalent increases in consumption calculated over the ergodic distribution of the model without bilateral loans, conditional on repayment.

The risk-inducing rule creates similar dynamics to the equilibrium with bargaining and, thus, welfare losses for the government. However, the size-dependent rule exemplifies the possibility of rules which induce lower default and spreads, improving the government’s welfare, while also generating profits for the large lender.

7. CONCLUDING REMARKS

Inspired by the rise of new bilateral lenders outside the Paris Club and with claims to de facto seniority relative to private creditors, we investigate the interaction between marketable debt and bilateral debt. We develop a model of sovereign debt in which the government can borrow from a large lender as well as from a competitive fringe of private creditors. We find that both sources of funds are linked by strong interactions, even when quantitatively the amounts borrowed from the large lender remain an order of magnitude smaller than market debts.

The model features a ‘relational overborrowing’ effect, which arises in the presence of the large lender. This effect is due to an endogenous cross-elasticity of bilateral borrowing terms to outcomes in debt markets, which erodes the market discipline of spreads. Our model of bargaining with the large lender generates this cross-elasticity with three key ingredients: loans from the large lender are costlier (or impossible) to default, their terms result from bargaining, and they are of shorter maturity than marketable debts. These ingredients capture the defining features of Central Bank swap lines, which [Horn et al. \(2021b\)](#) identify as a key instrument of senior bilateral sovereign lending. The simpler version of the model with exogenous bilateral terms shows that the cross-elasticity is in itself sufficient to incentivize the government to overborrow, face more sovereign risk, and ultimately attain lower levels of welfare.

The model with exogenous bilateral terms also illustrates a broader point. Policy discussions surrounding bilateral debts often focus on the price of the loans and on seniority itself. Our results emphasize that welfare losses are likely to come from a *different* source. In particular, we show how welfare losses can arise even when the government is free not to borrow from the bilateral lender if the terms offered are not advantageous enough relative to the market. We also provide examples of pricing schedules for the senior debt, like size-dependent rules, which improve welfare for the government without losses for the large lender. Rather, our main result is that the perverse incentives created by the cross-elasticity, which have been more overlooked, are crucial to generating the welfare losses we find. Our main model then describes a set of plausible assumptions, especially sequential bargaining over borrowing terms, under which the cross-elasticity could be expected to arise, thus creating perils of bilateral sovereign debt.

Our results suggest that increasing the amount of sources of indebtedness is not necessarily beneficial for the borrowing government. While bilateral loans can in some cases help a government fend off default, they can also make it more likely: either through reducing the effective costs of default or through the relational overborrowing effect.

The welfare impact of the large lender’s presence raises important policy questions and challenges. Limiting the use of bilateral debt during defaults is a clear welfare-enhancing policy in this model. Because the relational overborrowing effect operates through increased default risk, the gains of fiscal rules that constrain market borrowing should be larger for countries with access to the type of bilateral debts we describe.

The model implies a simple test to gauge the likely effects of a new bilateral creditor or instrument in practice. Bilateral loans whose interest rate is expected to be strongly decreasing in the amount (or spreads) of marketable debt will induce relational overborrowing and are thus likely to hurt welfare. More generally, it highlights the benefits of transparent rules for the terms of bilateral debts.

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A. MORE RESULTS

Increasing the borrower's bargaining power Figure 11 shows that when the borrower holds all the bargaining power, the loan interest rate is constant at β_L^{-1} . In this example with $\beta = \beta_L$, the upper bound on how large the loan m does not bind, but in the quantitative version (where market and bilateral debt coexist) with $\beta < \beta_L$, which at $\theta = 0$ recovers the model of [Hatchondo, Martinez, and Önder \(2017\)](#), the borrower prioritizes bilateral loans as a source of funding and thus quickly reaches the upper bound.

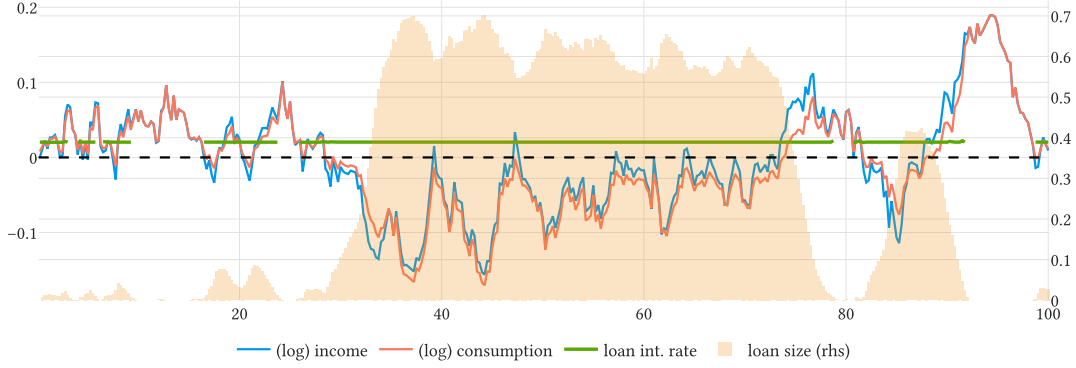


FIGURE 11: SIMULATED PATH, $\theta = 0$

Dynamics of loans around defaults Figure 12 shows that, further conditioning on an exclusion period of 2 years, the economy issues debt in the market in order to pay off the loan as soon as it recovers market access.

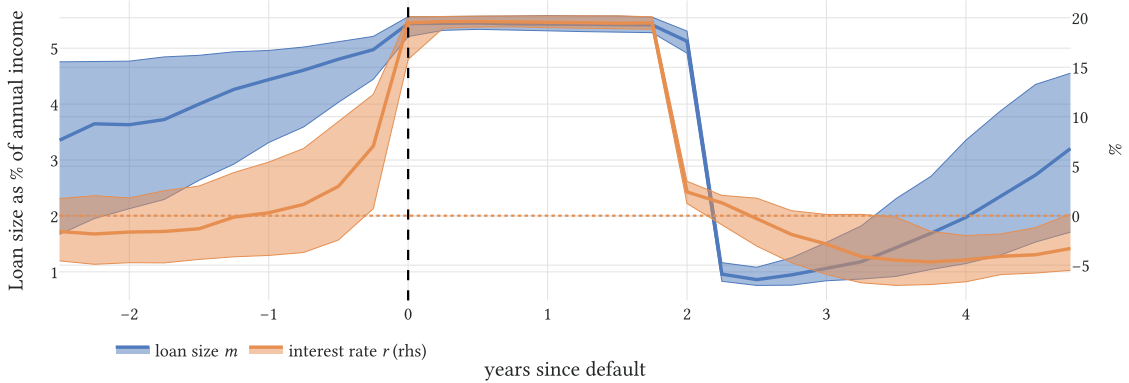


FIGURE 12: LOANS AROUND DEFAULT EVENTS

B. MODEL WITH BILATERAL LOANS ONLY

We study a simple model in which only bilateral loans are available. This model serves to clarify the dynamics of bargaining in bilateral lending and the strategy through which the large lender extracts surplus: subsidized terms while debt accumulates, combined with high interest rates when the debt stock becomes large and the borrower reduces leverage.

We model a small open economy borrowing from a large lender, who acts as a monopolist. The economy receives an endowment stream $y(z)$ where the state z follows an AR(1) process. Loans are short-term and therefore effectively continuously renegotiated. At the beginning of period t , let $v(m, z)$ represent the value attained by the government (or sovereign, or borrower) at income state z and owing m to the monopolist. Similarly, the large lender attains a value $h(m, z)$.

At the beginning of period t , borrower and lender negotiate over the terms of the loan. Payment of the full amount m extinguishes any debts and serves as a natural threat point. We use a simple Nash bargaining framework and set θ as the lender's bargaining power. The outcome of this negotiation is a transfer x and a new loan size m' determined by the solution to

$$\max_{x, m'} \mathcal{L}(x, m, m', z)^\theta \times \mathcal{B}(x, m, m', z)^{1-\theta} \quad (17)$$

where $\mathcal{L}$ and $\mathcal{B}$ represent the lender and borrower surplus functions, respectively. It will be useful to keep track of the implicit interest rate r of the loan which satisfies

$$x = \frac{1}{1+r} m' - m \quad (18)$$

After negotiations are concluded and transfers settled, consumption takes place. The large lender finances the net transfer x with a constant endowment a and thus consumes $c_L = a - x$. Conversely, the borrower receives the transfer so $c = y(z) + x$. Under risk neutral preferences for the lender,

$$\begin{aligned} \mathcal{L}(x, m, m', z) &= a - x + \beta_L \mathbb{E} [h(m', z') | z] - (a + m + \beta_L \mathbb{E} [h(0, z') | z]) \\ &= -x - m + \beta_L \mathbb{E} [h(m', z') - h(0, z') | z] \end{aligned}$$

and similarly

$$\mathcal{B}(x, m, m', z) = u(y(z) + x) - u(y(z) - m) + \beta \mathbb{E} [v(m', z') - v(0, z') | z]$$

where β_L and β are the discount factors of the lender and borrower, respectively. The borrower's utility function u is increasing and concave.

Notice that the choice of m' only affects continuation values v and h , while the choice of x only affects flow payoffs. Given a choice of m' , the first-order condition for x is

$$\mathcal{B}(x, m, m', z)\theta = \mathcal{L}(x, m, m', z)u'(y(z) + x)(1 - \theta)$$

Given the solution $\{x(m, z), m'(m, z)\}$ to (17), the value functions satisfy

$$\begin{aligned} v(m, z) &= u(y(z) + x(m, z)) + \beta \mathbb{E} [v(m'(m, z), z') \mid z] \\ h(m, z) &= a - x(m, z) + \beta_L \mathbb{E} [h(m'(m, z), z') \mid z] \end{aligned} \quad (19)$$

Finally, we normalize $a = 0$, which allows us to interpret $h(m, z)$ as the expected present discounted value of transfers along the equilibrium path, or the lender's total expected profits.

B.1 Equilibrium with bilateral loans only

We solve the model with bilateral loans only with a parametrization that illustrates the forces at play. Most parameters are set to standard values, Section 4 discusses calibration of the full version of the model with both types of debt. We choose $\theta = 0.5$ so the surplus is split equally between borrower and lender; we also set $\beta = \beta_L$ to isolate consumption smoothing and bargaining from the front-loading motive that would result if the borrower was relatively impatient, which in sovereign debt models tends to be the relevant case.

Figure 13 summarizes the terms of the new loan, for each level of income z and initial loan size m . Unsurprisingly, the borrower economy delevers in high-income states and receives positive transfers in low-income ones. The large lender makes intense use of the interest rate to extract surplus. When both debt and income are low, the lender offers subsidized and even negative rates. The benefit of incurring this cost is to induce high levels of debt, which make the borrower's future threat point more costly to exercise. Once the loan size is large, repaying it in full becomes difficult and the lender is able to charge much higher interest rates.

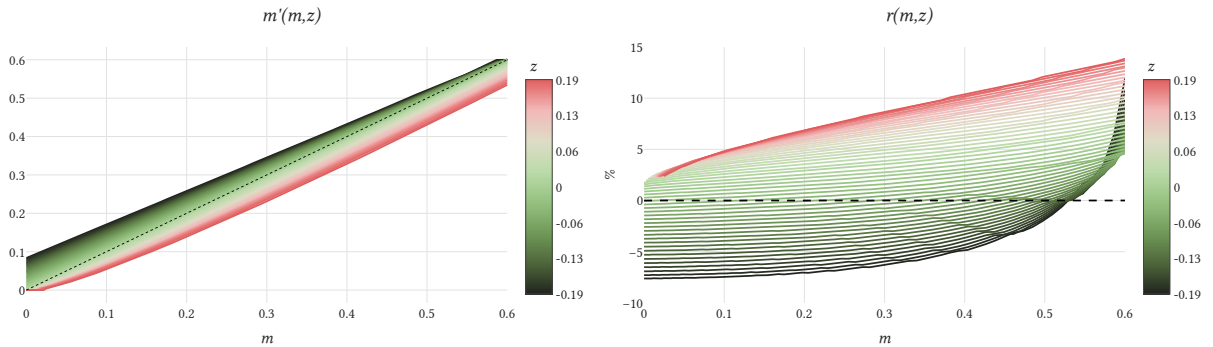


FIGURE 13: BILATERAL LENDING TERMS WITH $\theta = 0.5$

Figure 14 shows the value functions v and h for borrower and lender, respectively. As indebtedness m increases, the borrower's threat point becomes less credible. This increases the total surplus available, as new transfers are more valuable, but also makes the lender 'stronger' in the

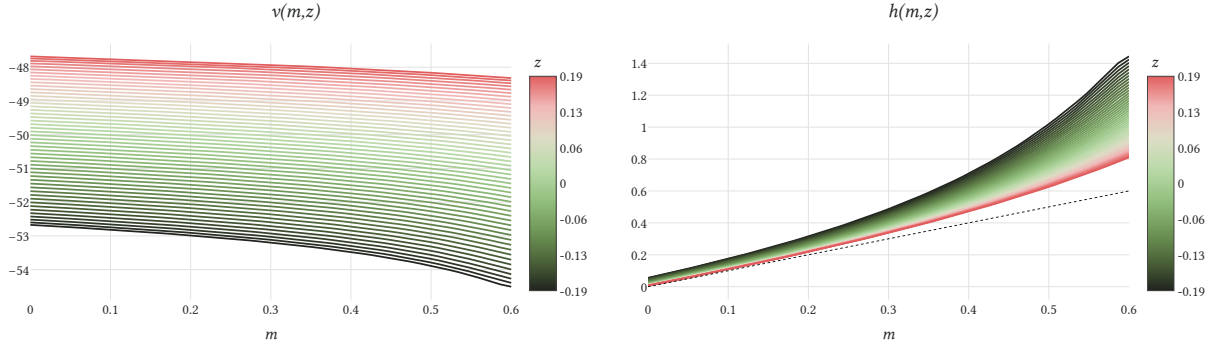


FIGURE 14: VALUE FUNCTIONS, $\theta = 0.5$

negotiation, leading to higher interest rates. This effect creates convexity in the lender's profits and, hence, in the value function h .

Convexity in the lender's value function implies endogenously risk-loving behavior. In equilibrium, the (risk-neutral) lender gambles for debt overhang. Subsidizing new loans in order to induce high indebtedness only pays off if the borrower's income takes a long time to recover. If the borrower receives a favorable income shock quickly, the loan is repaid before the lender has had an opportunity to raise rates and collect profits.

Figure 15 shows a simulation path, which further clarifies the lender's strategy. The loan is subsidized on the way up and, once debt has accumulated, the interest rate can increase to extract profits from the borrower. The borrower government anticipates these dynamics: the relationship between the initial subsidy and the expected high rates later on is disciplined by the requirement to deliver part of the surplus to the borrower, depending on the value of θ .

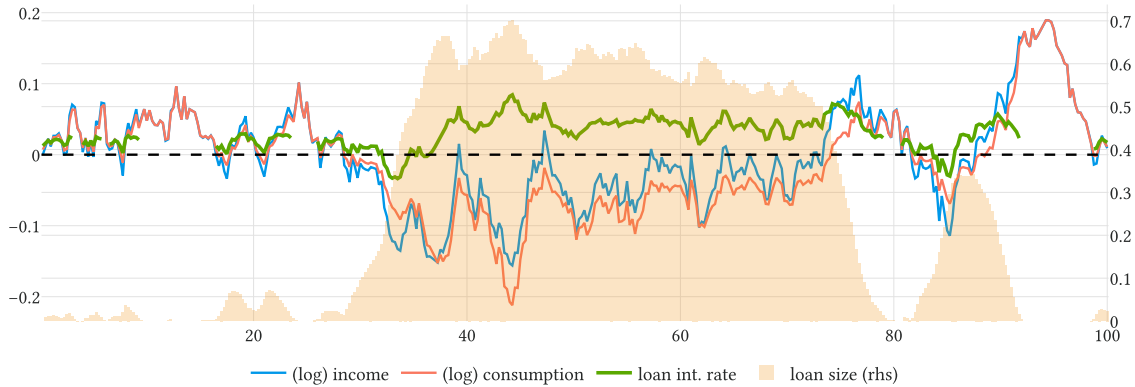


FIGURE 15: SIMULATED PATH, $\theta = 0.5$

Figure 11 in the Appendix, which simulates a model with $\theta = 0$, shows that when the borrower

holds all the bargaining power, it is able to borrow at rate β_L^{-1} at all times. Because rates do not go up once the loan is large, they cannot be negative when it is still small. This effectively recovers an income fluctuations problem at the risk-free rate without default.

The model with only bilateral loans available generates two main takeaways. First, bargained interest rates can increase strongly when the borrower's threat point is costly to exercise. In this version of the model this happens when income z is low and indebtedness m is high. Enforcement is not necessary for this result; even if it were possible to default on the loan, as long as the value of the threat point was decreasing in m , similar dynamics would obtain as the borrower government would be 'weak' in the same states. Second, anticipation of higher profits creates surplus for the lender and induces lower interest rates, which support accumulation of larger loan balances m . In the full model with both types of debt, marketable debt modulates the value of the government's threat point, as we describe below.